

CSE 425: Internet of Things

IoT Access Technologies

Courtesy: David Hanes and Co., Many Websites, and Google

Communications Criteria

Range: This section examines the importance of signal propagation and distance.

Frequency Bands: This section describes licensed and unlicensed spectrum, including sub-GHz frequencies.

Power Consumption: This section discusses the considerations required for devices connected to a stable power source compared to those that are battery powered.

Topology: This section highlights the various layouts that may be supported for connecting multiple smart objects.

Constrained Devices: This section details the limitations of certain smart objects from a connectivity perspective.

Constrained-Node Networks: This section highlights the challenges that are often encountered with networks connecting smart objects.

IoT Access Technologies

Standardization and alliances: The standards bodies that maintain the protocols for a technology

Physical layer: The wired or wireless methods and relevant frequencies

MAC layer: Considerations at the Media Access Control (MAC) layer, which bridges the physical layer with data link control

Topology: The topologies supported by the technology

Security: Security aspects of the technology

Competitive technologies: Other technologies that are similar and may be suitable alternatives to the given technology



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Connectivity Technologies – Part I

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Communication Protocols

The following communication protocols have immediate importance to consumer and industrial IoTs:

- ✓ IEEE 802.15.4
- ✓ Zigbee
- ✓ 6LoWPAN
- ✓ Wireless HART
- ✓ Z-Wave
- ✓ ISA 100
- ✓ Bluetooth
- ✓ NFC
- ✓ RFID



IEEE 802.15.4



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Features of IEEE 802.15.4

- ✓ Well-known standard for low data-rate WPAN.
- ✓ Developed for low-data-rate monitoring and control applications and extended-life low-power-consumption uses.
- ✓ This standard uses only the first two layers (PHY, MAC) plus the logical link control (LLC) and service specific convergence sub-layer (SSCS) additions to communicate with all upper layers
- ✓ Operates in the ISM band.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



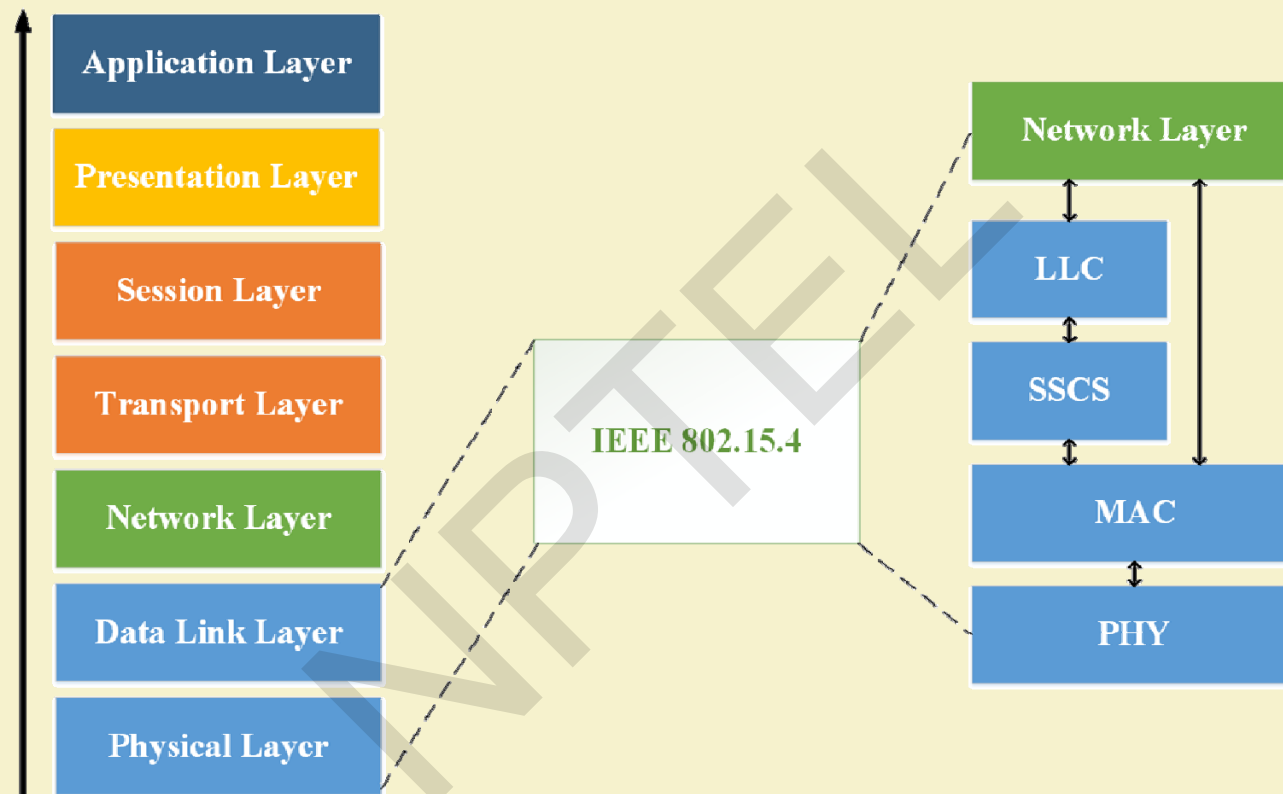
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- ✓ Uses direct sequence spread spectrum (DSSS) modulation.
- ✓ Highly tolerant of noise and interference and offers link reliability improvement mechanisms.
- ✓ Low-speed versions use Binary Phase Shift Keying (BPSK).
- ✓ High data-rate versions use offset-quadrature phase-shift keying (O-QPSK).
- ✓ Uses carrier sense multiple access with collision avoidance (CSMA-CA) for channel access.
- ✓ Multiplexing allows multiple users or nodes interference-free access to the same channel at different times.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



- ✓ Power consumption is minimized due to infrequently occurring very short packet transmissions with low duty cycle (<1%).
- ✓ The minimum power level defined is -3 dBm or 0.5 mW.
- ✓ Transmission, for most cases, is Line of Sight (LOS).
- ✓ Standard transmission range varies between 10m to 75m.
- ✓ Best case transmission range achieved outdoors can be upto 1000m.
- ✓ Networking topologies defined are -- Star, and Mesh.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



IEEE 802.15.4 Variants

A/B

- Base version

C

- For China

D

- For Japan

E

- Industrial applications

F

- Active RFID uses

G

- Smart utility networks (Smart Grids)



IEEE 802.15.4 Types

Networks

Non-Beacon
Enabled

Beacon
Enabled

(a)

802.15.4

FFD

RFD

(b)

PAN
coordinator

Router

Device

Device



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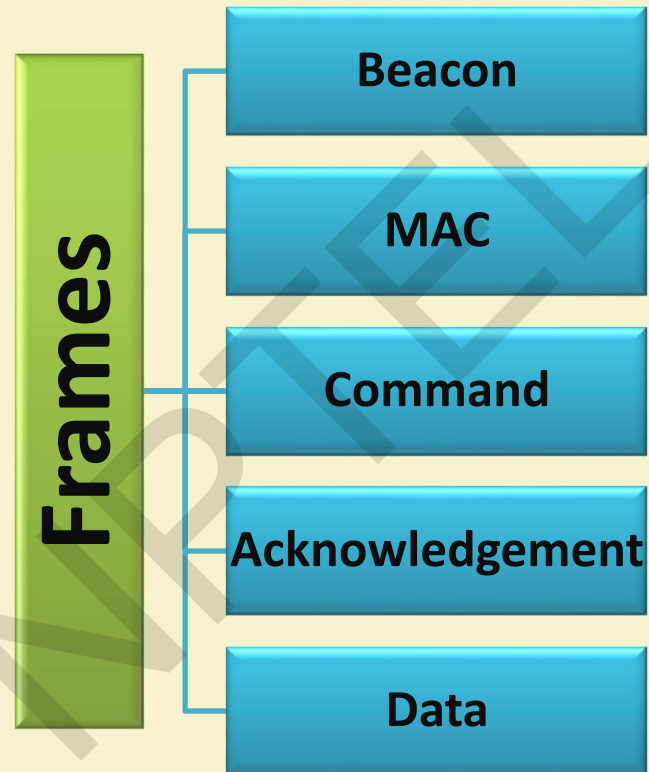
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- **Full Function Device (FFD)**
 - Can talk to all types of devices
 - Supports full protocol
- **Reduced Function Device (RFD)**
 - Can only talk to an FFD
 - Lower power consumption
 - Minimal CPU/RAM required



IEEE 802.15.4 Frames



Beacon Enabled Networks

- Periodic transmission of beacon messages
- Data-frames sent via Slotted CSMA/CA with a super frame structure managed by PAN coordinator
- Beacons used for synchronization & association of other nodes with the coordinator
- Scope of operation spans the whole network.



Non-Beacon Enabled Networks

- Data-frames sent via un-slotted CSMA/CA (Contention Based)
- Beacons used only for link layer discovery
- Requires both source and destination IDs.
- As 802.15.4 is primarily, a mesh protocol, all protocol addressing must adhere to mesh configurations
- De-centralized communication amongst nodes



Zigbee



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Features of ZigBee

- ✓ Most widely deployed enhancement of IEEE 802.15.4.
- ✓ The ZigBee protocol is defined by **layer 3 and above**. It works with the 802.15.4 layers 1 and 2.
- ✓ The standard uses layers 3 and 4 to define additional communication enhancements.
- ✓ These enhancements include authentication with valid nodes, encryption for security, and a data routing and forwarding capability that enables mesh networking.
- ✓ The most popular use of ZigBee is wireless sensor networks using the mesh topology.

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013





Important Components

ZDO

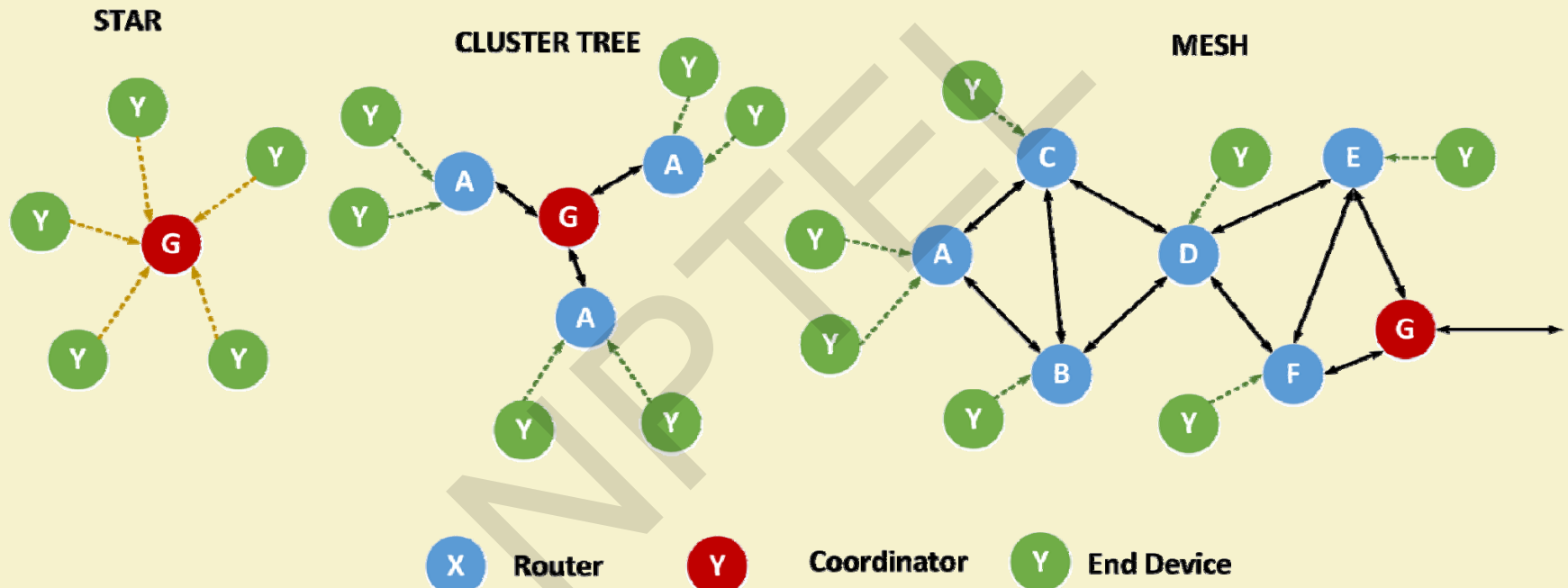
- ZigBee Device Object
(Device management, Security, Policies)

APS

- Application Support Sub-layer
(Interfacing and control services, bridge between network and other layers)



ZigBee Topologies



Source: T. Agarwal, "[ZigBee Wireless Technology Architecture and Applications](#)", Electronics Projects Focus (Online)



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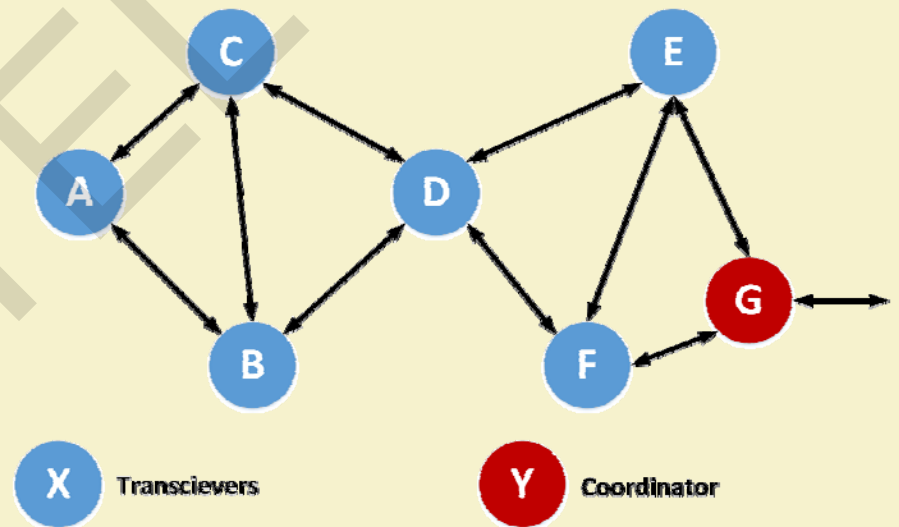


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ZigBee Mesh

- ✓ In a mesh, any node can communicate with any other node within its range.
- ✓ If nodes are not in range, messages are relayed through intermediate nodes.
- ✓ This allows the network deployment over large areas.



Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



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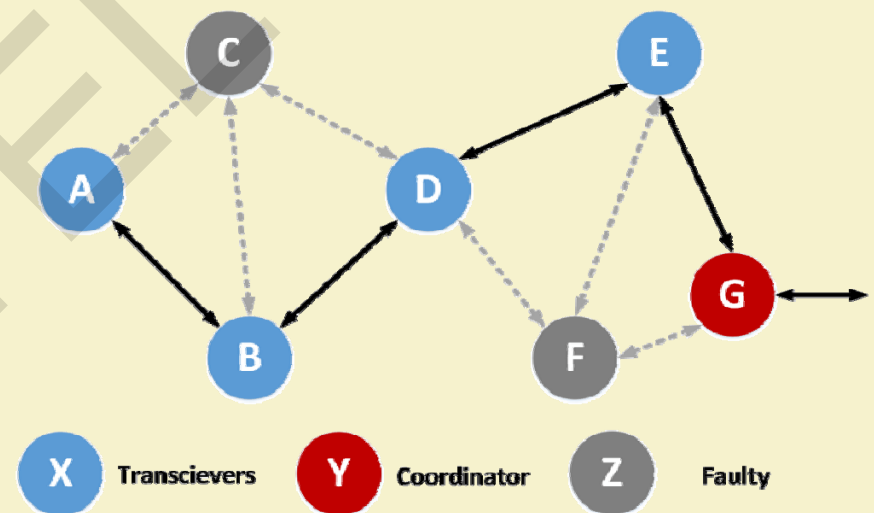


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ZigBee Mesh (Contd.)

- ✓ Meshes have increased network reliability.
- ✓ For example, if nodes C and F are down, the message packets from A can still be relayed to G via B and E.
- ✓ ZigBee mesh networks are self-configuring and self-healing.



Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



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ZigBee Types

✓ *ZigBee Coordinator (ZC):*

- The Coordinator forms the root of the ZigBee network tree and might act as a bridge between networks.
- There is a single ZigBee Coordinator in each network, which originally initiates the network.
- It stores information about the network under it and outside it.
- It acts as a Trust Center & repository for security keys.

Sources:

- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2010-04-15.
- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2009-02-05.



ZigBee Types

✓ *ZigBee Router (ZR):*

- Capable of running applications, as well as relaying information between nodes connected to it.

✓ *ZigBee End Device (ZED):*

- It contains just enough functionality to talk to the parent node, and it cannot relay data from other devices.
- This allows the node to be asleep a significant amount of the time thereby enhancing battery life.
- Memory requirements and cost of ZEDs are quite low, as compared to ZR or ZC.

Sources:

- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2010-04-15.
- "[Wireless Sensor Networks Research Group](#)". [Sensor-networks.org](#). 2009-02-05.



ZigBee Network Layer

- ✓ The network layer uses Ad Hoc On-Demand Distance Vector (AODV) routing.
- ✓ To find the final destination, the AODV broadcasts a route request to all its immediate neighbors.
- ✓ The neighbors relay the same information to their neighbors, eventually spreading the request throughout the network.
- ✓ Upon discovery of the destination, a low-cost path is calculated and informed to the requesting device via unicast messaging.

Source: ["Zigbee", Wikipedia \(Online\)](#)



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Applications

- ✓ Building automation
- ✓ Remote control (RF4CE or RF for consumer electronics)
- ✓ Smart energy for home energy monitoring
- ✓ Health care for medical and fitness monitoring
- ✓ Home automation for control of smart homes
- ✓ Light Link for control of LED lighting
- ✓ Telecom services

Source: L.Fenzel, [“What’s The Difference Between IEEE 802.15.4 And ZigBee Wireless?”](#), Electronic Design (Online), Mar. 2013



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Thank You!!



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